

MONODROMY OF GAUSS'S HYPERGEOMETRIC EQUATION

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1. INTRODUCTION

Gauss's Hypergeometric equation is a second-order linear differential equation given by:

$$\text{HG}_{a,b,c}(f) = z(1-z)\frac{d^2f}{dz^2} + [c - (a+b+1)z]\frac{df}{dz} - abf = 0 \quad 1.$$

From basic ODE theory, we know that the space of solutions to this ODE is a 2-dimensional $\mathbb{C}$ vector space. Additionally, we know that on any simply connected domain of $\mathbb{H}$, these solutions are holomorphic. One such distinguished solution is the Hypergeometric Function defined by the series

$$F(a, b, c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!} = 1 + \frac{ab}{c}z + \frac{a(a+1)b(b+1)}{c(c+1)} \frac{z^2}{2!} + \dots \quad 2.$$

where $(q)_n$ denotes the Pochhammer symbol, defined as

$$(q)_n = \begin{cases} 1 & \text{if } n = 0 \\ q(q+1)\dots(q+n-1) & \text{otherwise} \end{cases} \quad 3.$$

Our objective in this paper is to study the monodromy representation of the fundamental group of the $X = \mathbb{CP}^1 - \{3 \text{ points}\}$ acting on the solution space. Specifically, we will show that the analytic continuation of solutions around singular points gives rise to a representation

$$\rho : \pi_1(X) \rightarrow \text{PSL}_2(\mathbb{C}) \quad 4.$$

whose image can be described geometrically in terms of reflections across the sides of the Schwarz Triangle.

2. SINGULARITIES

To understand the monodromy, we must first classify the singularities of the equation.

Theorem 2.1 The hypergeometric equation is an ODE with three regular singular points on $\mathbb{CP}^1$ at $z = 0, 1$, and ∞ .

Proof We rewrite the hypergeometric equation in the standard form:

$$\frac{d^2f}{dz^2} + P(z)\frac{df}{dz} + Q(z)f = 0 \quad 5.$$

where:

$$\begin{aligned}
P(z) &= \frac{[c - (a + b + 1)z]}{z(1 - z)} \\
Q(z) &= -\frac{ab}{z(1 - z)}
\end{aligned} \tag{6}$$

By definition, a point z_0 is a regular singular point if $(z - z_0)P(z)$ and $(z - z_0)^2Q(z)$ are analytic at z_0 .

- At $z = 0$: $zP(z)$ and $z^2Q(z)$ are analytic.
- At $z = 1$: $(z - 1)P(z)$ and $(z - 1)^2Q(z)$ are analytic.

Thus, 0 and 1 are regular singular points.

To check $z = \infty$, we perform the coordinate change $w = \frac{1}{z}$. Then:

$$\begin{aligned}
\frac{df}{dz} &= \frac{df}{dw} \cdot \frac{dw}{dz} = -w^2 \frac{df}{dw} \\
\frac{d^2f}{dz^2} &= \frac{d}{dw} \left(-w^2 \frac{df}{dw} \right) \cdot \frac{dw}{dz} \\
&= \left(-2w \frac{df}{dw} - w^2 \frac{d^2f}{dw^2} \right) \cdot (-w^2) \\
&= w^4 \frac{d^2f}{dw^2} + 2w^3 \frac{df}{dw}
\end{aligned} \tag{7}$$

Substituting these into the original equation (and noting $z = \frac{1}{w}$):

$$w^4 \frac{d^2f}{dw^2} + 2w^3 \frac{df}{dw} + \frac{[c - \frac{a+b+1}{w}]}{\frac{1}{w}(1 - \frac{1}{w})} \left(-w^2 \frac{df}{dw} \right) - \frac{ab}{\frac{1}{w}(1 - \frac{1}{w})} f = 0 \tag{8}$$

After simplifying the coefficients, we divide by w^4 to normalize the leading term:

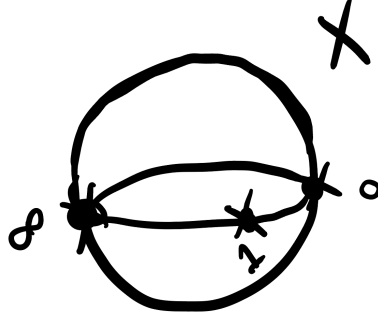
$$\frac{d^2f}{dw^2} + \left(\frac{2w - cw + a + b - 1}{w(w - 1)} \right) \frac{df}{dw} - \frac{ab}{w^2(w - 1)} f = 0 \tag{9}$$

Let the coefficients be $\tilde{P}(w)$ and $\tilde{Q}(w)$. We observe that $w\tilde{P}(w)$ and $w^2\tilde{Q}(w)$ are analytic at $w = 0$. Thus $w = 0$ (corresponding to $z = \infty$) is a regular singular point. $\square$

The set of singularities are $S = \{0, 1, \infty\}$. Thus our domain is

$$X = \mathbb{CP}^1 \setminus S \tag{10}$$

Let $x_0 \in X$ be a base point. By the standard existence and uniqueness theorem for linear ODEs, the space of solutions near x_0 , denoted V , is a complex vector space of dimension 2. Furthermore, the solutions are holomorphic on $\mathbb{H}$. We want to find a representation of $\pi_1(X) = \langle l_0, l_1 \rangle$ where l_0 is a nontrivial loop around 0 and l_1 is a nontrivial loop around 1.

FIGURE 1. Space $X = \mathbb{CP}^1 - \{0, 1, \infty\}$

3. THE BEHAVIOR OF INTEREST

Let $x_0 \in X$ be a base point. By the existence and uniqueness theorem for linear ODEs, the space of solutions near x_0 , denoted by V , is a complex vector space of dimension 2.

We are interested in the ratio of two independent solutions. Suppose F, G a basis of solutions. Consider the quotient map, $f : \mathbb{H} \rightarrow \mathbb{CP}^1$ where $f(z) = \frac{G(z)}{F(z)}$. If $\{F', G'\}$ is another basis for V , $\exists M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\mathbb{C})$ where

$$\begin{aligned} F'(z) &= dF(z) + cG(z) \\ G'(z) &= bF(z) + aG(z). \end{aligned} \tag{11}$$

The new ratio $f' = \frac{G'}{F'}$ is related to f by Möbius Transformation

$$\begin{aligned} f'(z) &= \frac{bF(z) + aG(z)}{dF(z) + cG(z)} \\ &= \frac{af(z) + b}{cf(z) + d} \end{aligned} \tag{12}$$

Then the action of $\pi_1(X)$ on V induces an action on the ratio f by Möbius transformations. This gives us the projective monodromy representation

$$\rho : \pi_1(X) \rightarrow \text{PGL}(V) \cong \text{PSL}_2(\mathbb{C}) \tag{13}$$

4. LOCAL BEHAVIOR OF SOLUTIONS NEAR SINGULARITIES

We want to understand the local behavior of solutions near each of the singularities. We will use the standard Frobenius method.

4.1. Near $z = 0$.

Assume a solution of the form

$$f(z) = z^\rho \sum_{n=0}^{\infty} a_n z^n \tag{14}$$

. The lowest order term of the ODE must vanish. Substituting the ansatz into the equation:

$$\begin{aligned}
0 &= (z - z^2)(a_0\rho(\rho - 1)z^{\rho-2} + \dots) + (c - (a + b + 1)z)(a_0\rho z^{\rho-1} + \dots) + ab(a_0z^\rho + \dots) \\
&= a_0[\rho(\rho - 1) + c\rho]z^{\rho-1} + O(z^\rho)
\end{aligned} \tag{15}$$

For $a_0 \neq 0$, the indicial equation is $\rho(\rho - 1) + c\rho = 0$, which simplifies to:

$$\rho(\rho - 1 + c) = 0 \Rightarrow \rho = 0 \text{ or } \rho = 1 - c \tag{16}$$

4.2. Near $z = 1$.

Let $w = z - 1$ be the local coordinate. The ODE transforms similarly. Focusing on the lowest order terms in w :

$$\begin{aligned}
0 &= -w(1 + w)(a_0\rho(\rho - 1)w^{\rho-2} + \dots) + [c - (a + b + 1)(w + 1)](a_0\rho w^{\rho-1} + \dots) + abf \\
&= a_0[-\rho(\rho - 1) + \rho(c - a - b - 1)]w^{\rho-1} + O(w^\rho)
\end{aligned} \tag{17}$$

The indicial equation is $\rho(c - a - b - 1 - \rho + 1) = 0$, which simplifies to:

$$\rho(c - a - b - \rho) = 0 \Rightarrow \rho = 0 \text{ or } \rho = c - a - b \tag{18}$$

4.3. Near $z = \infty$.

Using the coordinate $w = \frac{1}{z}$ and Equation 9:

$$\begin{aligned}
0 &= a_0[\rho(\rho - 1) + (1 - a - b)\rho + ab]w^{\rho-2} + O(w^{\rho-1}) \\
&= a_0[\rho^2 - \rho(a + b) + ab]w^{\rho-2} + O(w^{\rho-1})
\end{aligned} \tag{19}$$

The indicial equation is $\rho^2 - \rho(a + b) + ab = 0$, which factors as:

$$(\rho - a)(\rho - b) = 0 \Rightarrow \rho = a \text{ or } \rho = b \tag{20}$$

4.3.1. Summary of Local Solutions.

We can summarize the local exponents and the form of the canonical basis of solutions near each singularity. To define non integer powers like z^{1-c} we will take a branch cut along the negative imaginary axis (or a suitable ray in the lower half plane).

Singularity	Local Coordinate	Exponents (ρ)	Basis of Solutions F_{z_0}, G_{z_0}
0	z	0 $1 - c$	$F_0(z) = F(a, b, c; z)$ $G_0(z) = z^{1-c}\tilde{G}_0(z)$
1	$1 - z$	0 $c - a - b$	$F_1(z) = \text{Holomorphic at } z = 1$ $G_1(z) = (1 - z)^{1-c}\tilde{G}_0(1 - z)$
∞	$\frac{1}{z}$	a b	$F_\infty(z) = z^{-a}\tilde{F}_\infty(\frac{1}{z})$ $G_\infty(z) = z^{-b}\tilde{G}_\infty(\frac{1}{z})$

For the explicit construction of the monodromy group later, we will primarily work with the basis at $z = 0$:

$$V = \mathbb{C}\langle G_0, F_0 \rangle \tag{21}$$

Furthermore unless otherwise stated $F = F_0$ and $G = G_0$.

5. THE SCHWARZ TRIANGLE

In this section we will show that $f = \frac{G_0}{F_0}$ provides a conformal map between $\mathbb{H}$ to a curvilinear triangle, called the **Schwarz Triangle**. The triangle which gets mapped into can be seen in Figure 2. This geometric picture will be key to understanding the monodromy representation.

Additionally for notation, let

$$\begin{aligned}\alpha &= 1 - c \\ \beta &= c - a - b \\ \gamma &= b - a\end{aligned}\tag{22.}$$

We will assume that $0 < \alpha, \beta, \gamma < 1$. Let $f_0 = \frac{G_0}{F_0}$, $f_1 = \frac{G_1}{F_1}$, $f_\infty = \frac{G_\infty}{F_\infty}$. Unless otherwise specified $f = f_0$. As established in Section 3, these functions are related by Möbius transformations. Note f is holomorphic on $\mathbb{H}$, because both F_0 and G_0 are holomorphic on $\mathbb{H}$ and for our choice of parameters F_0 has no zeroes on $\mathbb{H}$.

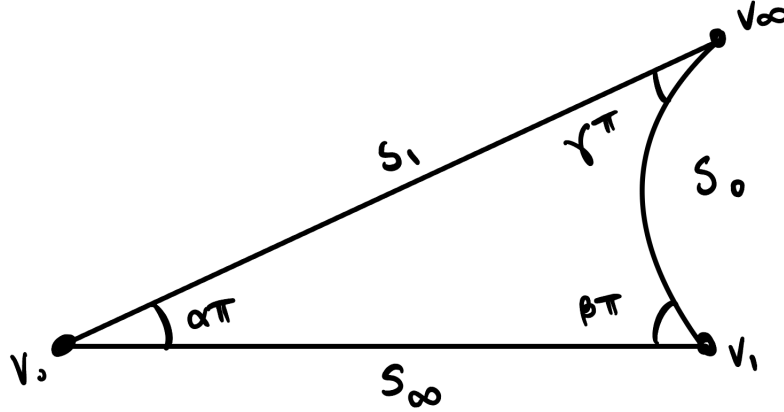


FIGURE 2. Schwarz Triangle

Theorem 5.1 The function

$$f = \frac{G_0}{F_0} : \mathbb{H} \rightarrow T\tag{23.}$$

where T is a curvilinear triangle with vertices $v_0 = f(0)$, $v_1 = f(1)$ and $v_\infty = f(\infty)$ and interior angles $\pi\alpha$, $\pi\beta$, $\pi\gamma$ respectively. The triangle which gets mapped into can be seen in Figure 2.

Proof F_0 is real on the real axis and G_0 is real on the positive real axis. Since the branch cut is made along a ray in the lower half plane, $\forall z \in (-\infty, 0)$, $\arg(z) = \pi$. Thus $\forall z \in (-\infty, 0)$, $z = |z| e^{i\pi}$. Thus

$$\begin{aligned}G_0(z) &= z^\alpha \tilde{G}_0(z) \\ &= (|z| e^{i\pi})^\alpha \tilde{G}_0(z) \\ &= |z|^{1-c} e^{i\pi\alpha} \tilde{G}_0(z)\end{aligned}\tag{24.}$$

thus $G_0((-\infty, 0)) \subset \{re^{i\pi\alpha} : r \in \mathbb{R}\}$. Moreover it is known that $\tilde{G}_0(z)$ is strictly positive. Thus $G_0((-\infty, 0)) \subset \{re^{i\pi\alpha} : r \in (0, \infty)\}$. Thus f_0 maps $S = \{z \in \mathbb{H} : |z| < 1\}$ to the wedge shaped region $\{w : 0 < \arg w < \pi\alpha\}$. Using a similar argument, we can see f_1 maps the S to $\{w : 0 < \arg w < \pi\beta\}$ and f_∞ maps S to $\{w : 0 < \arg w < \pi\gamma\}$. However we are interested in the behavior of a single global function $f = f_0$. We know that near $z = 1$, $\exists M \in \text{PSL}_2(\mathbb{C})$ where

$$f(z) = M(f_1(z)) \tag{25}$$

where M acts by Möbius transformation. Möbius transformations have the following key geometric properties:

1. Conformality: They preserve angles
2. Generalized Circles: They map circlines to circlines.

Since f is holomorphic on $\mathbb{H}$ and continuous on $\mathbb{R} - \{0, 1\}$, the boundary of the image domain is determined by the image of $\mathbb{R} \cup \{\infty\}$. We can analyze the image of the segments of the three segments of the real line:

1. The segment $(0, 1)$: Since F and G have real coefficients, then for $z \in (0, 1)$ $F(z), G(z) \in \mathbb{R}$. Consequently $f(z)$ maps the interval to a segment of the real line.
2. The segment $(-\infty, 0)$: We know that near $z = 0$, f_0 maps this segment to a ray with $\arg z = \pi\alpha$. Since $f = f_0$, the image of $(-\infty, 0)$ is a straight line meeting the image of $(0, 1)$ at the vertex $f(0)$ with interior angle $\pi\alpha$
3. The segment $(1, \infty)$: Near $z = 1$, the local function f_1 maps the boundary to a wedge of angle $\pi\beta$. Since $f = M \circ f_1$ and M maps the straight sides of f_1 -wedge to circlines, f maps the segment $(1, \infty)$ to a circline. Because M is conformal, the circline must meet $f((0, 1))$ at v_1 with internal angle $\pi\beta$
4. The vertex at ∞ : Similarly, near $z = \infty$, the relation between f and f_∞ implies that the arcs coming from $(-\infty, 0)$ and $(1, \infty)$ meet the vertex v_∞ with angle at $\pi\gamma$

Thus f maps $\mathbb{H}$ to a curvilinear triangle with vertices v_0 , v_1 , and v_∞ .

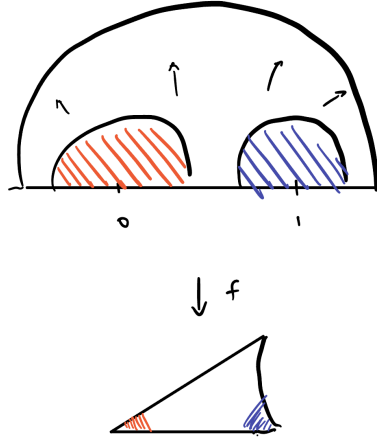


FIGURE 3. Mapping semicircles and then stitching

□

Theorem 5.2 $f : \mathbb{H} \rightarrow T$ is a conformal bijection.

Proof

We will first show that $f'(z) \neq 0$ on $\mathbb{H}$ to get local injectivity and then use the argument principle to get global injectivity. Let $W(z) = G'(z)F(z) - G(z)F'(z)$. Thus

$$f'(z) = \frac{W(z)}{(F(z))^2}. \quad 26.$$

Differentiating $W(z)$, we see

$$\begin{aligned} \frac{dW}{dz} &= \frac{d^2G}{dz^2}F + \frac{dG}{dz}\frac{dF}{dz} - \frac{dG}{dz}\frac{dF}{dz} - G\frac{d^2F}{dz^2} \\ &= \frac{d^2G}{dz^2}F - G\frac{d^2F}{dz^2}. \end{aligned} \quad 27.$$

Since Both F and G are solutions to $\text{HG}_{a,b,c}(f) = 0$. We know that

$$\begin{aligned} \frac{d^2G}{dz^2} &= \frac{1}{z(z-1)} \left[(c - (a+b+1)z) \frac{dG}{dz} - abG \right] \\ \frac{d^2F}{dz^2} &= \frac{1}{z(z-1)} \left[(c - (a+b+1)z) \frac{dF}{dz} - abF \right] \end{aligned} \quad 28.$$

Plugging into $\frac{dW}{dz}$ and simplifying:

$$\begin{aligned}
\frac{dW}{dz} &= \frac{1}{z(z-1)} \left[(c - (a+b+1)z) \left(F \frac{dG}{dz} - G \frac{dF}{dz} \right) - abGF + abGF \right] \\
&= \frac{1}{z(z-1)} \left[(c - (a+b+1)z) \left(F \frac{dG}{dz} - G \frac{dF}{dz} \right) \right] \\
&= \frac{1}{z(z-1)} (c - (a+b+1)z) W
\end{aligned} \tag{29}$$

Thus W is the solution to the following first order ODE,

$$z(1-z) \frac{dW}{dz} + (c - (a+b+1)z)W = 0 \tag{30}$$

We can explicitly solve for W .

$$\frac{dW}{W} = - \left[\frac{c - (a+b+1)z}{z(1-z)} \right] dz \tag{31}$$

Solving this differential equation we get:

$$W(z) = Cz^{-c}(1-z)^{a+b+1-c} = Ce^{-c \log(z)} e^{(a+b+1-c) \log(1-z)} \tag{32}$$

where $C \neq 0$ (F and G are independent solutions). Note the following that $\log(z)$ is well defined for $z \in (\mathbb{H} \cup \mathbb{R}) - \{0, 1\}$ because the branch cut is in the lower half plane. Thus W is both well defined and nonzero on $(\mathbb{H} \cup \mathbb{R}) - \{0, 1\}$. This implies that $f'(z)$ is nonzero and thus a local homeomorphism. Furthermore, on $\mathbb{R} - \{0, 1\}$, f' is monotonic. Thus it is injective on $\mathbb{R} - \{0, 1\}$.

To prove injectivity, we will use the argument principle. Pick $w_0 \in T$. Let $\Gamma_{R,\varepsilon}$ be a indented semicircle contour of radius R centered at the origin, with a indent of radii ε at the origin. Since f is holomorphic on $\mathbb{H}$, it has no poles in $\mathbb{H}$. By the argument principle,

$$\frac{1}{2\pi i} \lim_{\substack{R \rightarrow \infty \\ \varepsilon \rightarrow 0}} \int_{\Gamma_{R,\varepsilon}} \frac{f'(z)}{f(z) - w_0} dz = N \tag{33}$$

where N are the number of zeros of $f(z) - w_0$. The integral also has a topological interpretation as the winding number of $\Gamma = f(\partial\mathbb{H})$, the image curve, around the point w_0 . As previously shown, $f(\partial\mathbb{H})$ is the boundary of the triangle, which we have shown gets mapped injectively. Thus Γ is a simple closed curve and has a winding number of 1. Thus $N = 1$. Thus $f : \mathbb{H} \rightarrow T$ is a conformal bijection. $\square$

6. MONODROMY REPRESENTATION

Using the geometric picture of the Schwarz triangle we compute $\rho(\pi_1(X)) \subset \text{PSL}_2(\mathbb{C})$. We begin with a technical lemma about reflections.

Lemma 6.1 Let $\mathcal{L}$ be the set of all circlines in $\mathbb{CP}^1$. For any circline $c \in \mathcal{L}$, let $r_c : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ denote reflection across c . Then r_c is antiholomorphic.

Proof Möbius transformations in $\text{PSL}_2(\mathbb{C})$ act holomorphically and transitively on ordered triples of distinct points in $\mathbb{CP}^1$. Since any circline is uniquely determined by three distinct points on it, and $\mathbb{RP}^1 \subset \mathbb{CP}^1$ is a circline, it follows that Möbius transformations act transitively on circlines.

Therefore, for any circline $c \in L$, there exists $M_c \in \text{PSL}_2(\mathbb{C})$ such that $M_{c(c)} = \mathbb{RP}^1$. Reflection across c can be expressed as

$$r_{c(z)} = M_c^{-1} \circ r_{\mathbb{RP}^1} \circ M_c \quad 34.$$

where $r_{\mathbb{RP}^1}(z) = \bar{z}$ is reflection across the real axis.

Since $r_{\mathbb{RP}^1}(z) = \bar{z}$ satisfies $\frac{\partial \bar{z}}{\partial z} = 0$, it is antiholomorphic. The composition $r_{\mathbb{RP}^1} \circ M_c$ of an antiholomorphic function with a holomorphic function is antiholomorphic. Finally, $r_c = M_c^{-1} \circ (r_{\mathbb{RP}^1} \circ M_c)$ is the composition of a holomorphic function M_c^{-1} with an antiholomorphic function, which is antiholomorphic. $\square$

The next theorem is the technical foundation for computing monodromy.

Theorem 6.2 Let $f : \mathbb{H} \rightarrow T$ be the conformal map defined in Theorem 5.2, from upper half-plane to the Schwarz triangle. Suppose f extends continuously to a real interval $[a, b] \in \mathbb{R} \cup \infty$, mapping it to a side s of T . Let $r_s : \mathbb{CP}^1 \rightarrow \mathbb{CP}^1$ denote reflection across the side s .

Define $g : \mathbb{H}^- \rightarrow \mathbb{CP}^1$ by

$$g(z) = r_s(f(\bar{z})) \quad 35.$$

where $\mathbb{H}^- = \{z \in \mathbb{C} : \Im(z) < 0\}$ is the lower half-plane. Then g is holomorphic and provides an analytic continuation of f to $\mathbb{H}^-$ across the interval $[a, b]$.

Proof r_s is antiholomorphic by Lemma 6.1. $z \rightarrow f(\bar{z})$ is a composition of a holomorphic and antiholomorphic function and is thus antiholomorphic. Thus g is holomorphic, because it is the composition of a antiholomorphic function with a antiholomorphic one.

Note that $x \in (a, b)$, f extends continuously to $[a, b]$ and maps it to the side s , so $f(x) \in s$. Reflection across a side fixes the points on the side, thus $g(x) = r_s(f(\bar{x})) = r_s(f(x)) = f(x)$.

Let $F : \mathbb{H}^+ \cup (a, b) \cup \mathbb{H}^- \rightarrow \mathbb{CP}^1$ where

$$F(z) := \begin{cases} f(z) & \text{if } z \in \mathbb{H}^+ \cup (a, b) \\ g(z) & \text{if } z \in \mathbb{H}^- \end{cases} \quad 36.$$

Let γ be a any closed curve on $\mathbb{H}^+ \cup (a, b) \cup \mathbb{H}^-$.

1. $\gamma \subset \mathbb{H}^+$: Since f is holomorphic,

$$\int_{\gamma} F = \int_{\gamma} f = 0 \quad 37.$$

2. $\gamma \subset \mathbb{H}^-$: Since g is holomorphic,

$$\int_{\gamma} F = \int_{\gamma} g = 0 \quad 38.$$

3. Otherwise: Split γ into pieces in $\mathbb{H}^+$ and $\mathbb{H}^-$ respectively that approach the boundary (a, b) . F is continous and holomorphic on each peice, and the integrals along (a, b) from opposite sides cancel (since f and g agree there), we have $\int_{\gamma} f = 0$

By Morera's Theorem, F is holomorphic on $\mathbb{H}^+ \cup (a, b) \cup \mathbb{H}^-$. $\square$

We now use Theorem 6.2 to determine the effect of loops around each singular point.

Theorem 6.3 (*Monodromy Representation via reflection*) Let T be the Schwarz triangle defined above with vertices $v_0 = f(0)$, $v_1 = f(1)$, $v_\infty = f(\infty)$ and interior angles $\pi\alpha$, $\pi\beta$, and $\pi\gamma$ respectively. Let s_0, s_1, s_∞ be the sides opposite to v_0, v_1, v_∞ respectively. Let R_0 be a rotation of $2\pi\alpha$ about v_0 and R_1 be a rotation of $2\pi\beta$ about v_1 . The monodromy representation $\rho : \pi_1(X) \rightarrow \text{PSL}_2(\mathbb{C})$ is given by

$$\begin{aligned}\rho(l_0) &= r_{s_1} \circ r_{s_\infty} = R_0 \\ \rho(l_1) &= r_{r_\infty(s_0)} \circ r(s_\infty) = R_1\end{aligned}\tag{39}$$

Proof We will analyze the effect of analytic continuation of $f = \frac{G_0}{F_0}$ around the generators of $\pi_1(X)$. By Theorem 5.2, $f : \mathbb{H} \rightarrow T$ is a conformal bijection where

1. f maps $(0, 1)$ to the side s_∞
2. f maps $(1, \infty)$ to the side s_0
3. f maps $(-\infty, 0)$ to the side s_1

By Theorem 6.2, analytic continuation across a boundary segment corresponds to reflection of the image across the corresponding side of the triangle.

Analytic continuation around l_0 : Let l_0 be a small loop based at $z_0 \in \mathbb{H}$ that loops around 0 once counter clockwise. Thus l_0 crosses $(-\infty, 0)$ into $\mathbb{H}^-$ and then crosses $(0, 1)$ into $\mathbb{H}^+$. By Theorem 6.2, analytic continuation of f into $\mathbb{H}^-$ across $(-\infty, 0)$ is given by $f^{(1)}(z) = r_{s_1}(f(\bar{z}))$. The image of $\mathbb{H}^-$, is given by reflecting the triangle T across the side s_1 , which we will denote as $T' = r_{s_1}(T)$. Continuing along l_0 , we cross back from $\mathbb{H}^-$ to $\mathbb{H}$ through the interval $(0, 1)$. To continue $f^{(1)}$ across $(0, 1)$, note that f^1 maps $(0, 1)$ (now approached from below) to $r_{s_1}(s_\infty)$. By Theorem 6.2, continuation across this boundary is given by $f^{(2)}(z) = r_{r_{s_1}(s_\infty)}(f^{(1)}(\bar{z})) = r_{r_{s_1}(s_\infty)}(r_{s_1}(f(z)))$. By the geometry of reflections, $r_{r_{s_1}(s_\infty)} = r_{s_1} \circ r_{s_\infty} \circ r_{s_1}$. This is because you first transform the reflected circline back to the original circline, reflect here, and then transform back. Thus

$$f^{(2)}(z) = (r_{s_1} \circ r_{s_\infty} \circ r_{s_1})(r_{s_1}(f(z))) = (r_{s_1} \circ r_{s_\infty})(f(z)).\tag{40}$$

Thus traveling a l_0 sends $f \rightarrow r_{s_1} \circ r_{s_\infty} \circ f$. Thus $\rho(l_0) = r_{s_1} \circ r_{s_\infty}$. Furthermore, by the geometry of rotations and reflections, $r_{s_1} \circ r_{s_\infty} = R_0$. Thus $\rho(l_0) = R_0$.

We will do a similar argument for the analytic continuation around l_1 .

Let l_1 be a small loop based at $z_0 \in \mathbb{H}$ that loops around 1 once counter clockwise. Thus l_1 crosses $(0, 1)$ into $\mathbb{H}^-$ and then crosses $(1, \infty)$ into $\mathbb{H}^+$. By Theorem 6.2, analytic continuation of f into $\mathbb{H}^-$ across $(0, 1)$ is given by $f^{(1)}(z) = r_{s_\infty}(f(\bar{z}))$. The image of $\mathbb{H}^-$, is given by reflecting the triangle T across the side s_∞ , which we will denote as $T' = r_{s_\infty}(T)$. Continuing along l_1 , we cross back from $\mathbb{H}^-$ to $\mathbb{H}$ through the interval $(1, \infty)$. To continue $f^{(1)}$ across $(1, \infty)$, note that f^1 maps $(1, \infty)$ (now approached from below) to $r_{s_\infty}(s_0)$. By Theorem 6.2, continuation across this boundary is given by $f^{(2)}(z) = r_{r_{s_\infty}(s_0)}(f^{(1)}(\bar{z})) = r_{r_{s_\infty}(s_0)}(r_{s_\infty}(f(z)))$. By the geometry of reflections, $r_{r_{s_\infty}(s_0)} = r_{s_\infty} \circ r_{s_0} \circ r_{s_\infty}$. This is because you first transform the reflected circline back to the original circline, reflect here, and then transform back. Thus

$$f^{(2)}(z) = (r_{s_\infty} \circ r_{s_0} \circ r_{s_\infty})(r_{s_\infty}(f(z))) = (r_{s_\infty} \circ r_{s_0})(f(z)). \quad 41.$$

Thus traveling a l_1 sends $f \rightarrow r_{s_\infty} \circ r_{s_0} \circ f$. Thus $\rho(l_1) = r_{s_\infty} \circ r_{s_0}$. Furthermore, by the geometry of rotations and reflections, $r_{s_\infty} \circ r_{s_0} = R_1$. Thus $\rho(l_0) = R_1$.

Let l_∞ be the loop that loops around infinity. Since $\pi_1(X) = \langle l_0, l_1 \rangle$, let l_∞ be oriented such that

$$l_\infty = l_0 l_1 \in \pi_1(X) \quad 42.$$

. Then

$$\rho(l_\infty) = \rho(l_0)\rho(l_1) = r_{s_1} \circ r_{s_\infty} \circ r_{s_\infty} \circ r_{s_0} = r_{s_1} \circ r_{s_0}. \quad 43.$$

This is a rotation by $2\pi\gamma$ around v_∞ . $\square$

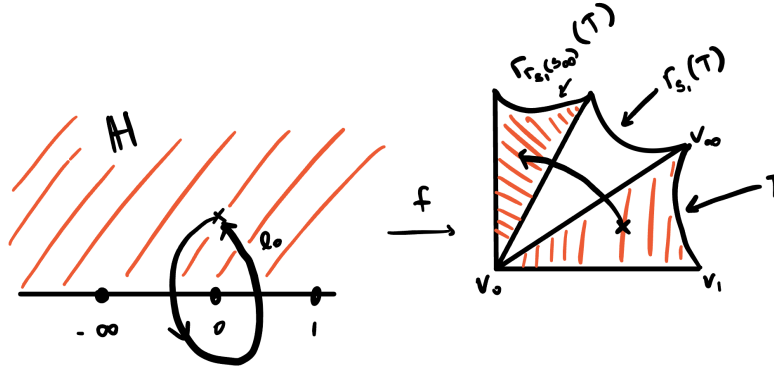


FIGURE 4. Behavior as we traverse l_0

Corollary 6.4 $\rho(\pi_1(X)) \subset \text{PSL}_2(\mathbb{C})$ is a group generated by rotations by angles $2\pi\alpha$, $2\pi\beta$, and $2\pi\gamma$ about the vertices of the Schwarz Triangle. When $\alpha = \frac{1}{p}$, $\beta = \frac{1}{q}$, and $\gamma = \frac{1}{r}$ for positive p, q, r . $\rho(\pi_1(X)) = \Delta(p, q, r)$ the triangle group.

6.0.1. Explicit Matrices.

We will explicitly calculate $\rho(\pi_1(X))$ under a basis of the action close to v_0 . l_0 is a loop around 0, $F_0(ze^{2\pi i}) = F_0(z)$ where as $G_0(ze^{2\pi i}) = e^{2\pi i\alpha}G_0(z)$. Thus analytically continuing f around 0, sends $f \rightarrow e^{2\pi i\alpha}f$. This means that

$$\rho(l_0) = \begin{pmatrix} e^{2\pi i\alpha} & 0 \\ 0 & 1 \end{pmatrix}. \quad 44.$$

Furthermore, we know that $\rho(l_1)$ is a rotation around v_1 . That means under this basis,

$$\begin{aligned} \rho(l_1) &= \begin{pmatrix} 1 & v_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{2\pi i\beta} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -v_1 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} e^{2\pi i\beta} & v_0 + v_0 e^{2\pi i\beta} \\ 0 & 1 \end{pmatrix} \end{aligned} \quad 45.$$

From the classical theory of hypergeometric functions (Whittaker & Watson), the connection formula gives:

$$v_1 = f(1) = \frac{\Gamma(2-c)\Gamma(c-a)\Gamma(c-b)}{\Gamma(c)\Gamma(1-a)\Gamma(1-b)}. \quad 46.$$

7. APPENDIX: EXPLICIT CONSTRUCTION OF SOLUTION BASES

In this section, we explicitly derive the basis solutions near each singularity by transforming the hypergeometric equation. The main technique is the Frobenius method, where we substitute the ansatz $f(z) = z^r \tilde{f}(z)$.

Let f be a solution to $\text{HG}_{a,b,c}(f) = 0$. We wish to find a solution of the form $f(z) = z^r \tilde{f}(z)$. Computing derivatives

$$\begin{aligned} f'(z) &= rz^{r-1} \tilde{f}(z) + z^r \tilde{f}'(z) \\ f''(z) &= r(r-1)z^{r-2} \tilde{f}(z) + 2rz^{r-1} \tilde{f}'(z) + z^r \tilde{f}''(z). \end{aligned} \quad 47.$$

Plugging f , f' , and f'' into $\text{HG}_{a,b,c}(f)$ we see that,

$$\begin{aligned} 0 &= z(1-z)f'' + [c - (a+b+1)z]f' - abf \\ &= z^{r+1}(1-z)\tilde{f}'' + [2rz^r(1-z) + z^r(c - (a+b+1)z)]\tilde{f}' \\ &\quad - [r(r-1)z^{r-1}(1-z) + rz^{r-1}(c - (a+b+1)z) + z^r]\tilde{f} \\ &= z^{r+1}(1-z)\tilde{f}'' + z^r[2r(1-z) + (c - (a+b+1)z)]\tilde{f}' \\ &\quad + z^{r-1}[r(r-1)(1-z) + r(c - (a+b+1)z) - abz]\tilde{f} \end{aligned} \quad 48.$$

7.0.1. At $z = 0$.

We have two values for r , $r = 0$ and $r = 1 - c$. $r = 0$ gives us the standard Hypergeometric Equation back. But for $r = 1 - c$, we see our 2nd order differential equation for $\tilde{f}$ transforms. We will take this coefficient by coefficient.

$$\begin{aligned} z^{1-c}[2(1-c)(1-z) + (c - (a+b+1)z)] &= z^{1-c}(2 - c - 3z - az - bz + 2cz) \\ &= z^{1-c}[(2-c) - (a+b-2c+2+1)z] \end{aligned} \quad 49.$$

and

$$z^{-c}[(1-c)(-c)(1-z) + (1-c)(c - (a+b+1)z) - abz] = -z^{1-c}a'b' \quad 50.$$

where $a' = a - c + 1$ and $b' = b - c + 1$. Let $c' = 2 - c$. Thus $\text{HG}_{a',b',c'}(\tilde{f}) = 0$ and thus $\tilde{f}(z) = F(a - c + 1, b - c + 1, 2 - c; z)$. Thus

$$\begin{aligned} F_0(z) &= F(a, b, c; z) \\ G_0(z) &= z^{1-c}F(a - c + 1, b - c + 1, 2 - c; z) \end{aligned} \quad 51.$$

7.0.2. At $z = 1$:

We will first transform our standard hypergeometric equation $\text{HG}_{a,b,c}(f)$ with the $w = 1 - z$ variable transformation and study how the differential equation changes.

$$\begin{aligned}\frac{d}{dz} &= -\frac{d}{dw} \\ \frac{d}{dz^2} &= \frac{d}{dw^2}\end{aligned}\tag{52}$$

Thus,

$$w(1-w)\frac{d^2f}{dw^2} + [(a+b+1)(1-w) - c]\frac{df}{dw} - abf = 0\tag{53}$$

We will specifically study,

$$(a+b+1)(1-w) - c = (a+b+1-c) - w(a+b+1)\tag{54}$$

Let $a' = a$, $b' = b$, and $c' = a+b+1-c$. Thus

$$\text{HG}_{a',b',c'}(f) = w(1-w)\frac{d^2f}{dw^2} + [c' - w(a' + b' + 1)]\frac{df}{dw} + a'b'f = 0\tag{55}$$

By the Frobenius method we know the two solutions of $\text{HG}_{a',b',c'}(f)$ at $w = 0$,

$$\begin{aligned}f_1(w) &= F(a', b', c'; w) \\ f_2(w) &= w^{1-c'}F(a' - c' + 1, b' - c' + 1, 2 - c'; w) \\ &= w^{c-a-b}F(c-b, c-a, c-a-b+1; w)\end{aligned}\tag{56}$$

Thus,

$$\begin{aligned}F_1(w) &= F(a, b, c; z) \\ G_2(z) &= (1-z)^{c-a-b}F(c-b, c-a, c-a-b+1; 1-z)\end{aligned}\tag{57}$$

7.0.3. $z = \infty$.

Using Equation 9, we have

$$\frac{d^2f}{dw^2} + \left(\frac{2w - cw + a + b - 1}{w(w-1)}\right)\frac{df}{dw} - \frac{ab}{w^2(w-1)}f = 0\tag{58}$$

thus,

$$w^2(w-1)\frac{d^2f}{dw^2} + w(2w - cw + a + b - 1)\frac{df}{dw} - abf = 0\tag{59}$$

We have to use the fact that we know that $f(w) = w^a \tilde{f}(w)$ for some $\tilde{f}(w)$. We will solve for $\tilde{f}$ and then by symmetry we should also know what $g(w) = w^b \tilde{g}(w)$ is.

$$\begin{aligned}f'(w) &= aw^{a-1}\tilde{f}(w) + w^a\tilde{f}'(w) \\ f''(w) &= a(a-1)w^{a-2}\tilde{f}(w) + 2aw^{a-1}\tilde{f}'(w) + w^a\tilde{f}''(w)\end{aligned}\tag{60}$$

Thus

$$\begin{aligned}0 &= w^2(w-1)\left(a(a-1)w^{a-2}\tilde{f} + 2aw^{a-1}\frac{d\tilde{f}}{dw} + w^a\frac{d^2\tilde{f}}{dw^2}\right) \\ &+ w(2w - cw + a + b - 1)\left(aw^{a-1}\tilde{f}(w) + w^a\frac{d\tilde{f}}{dw}\right) \\ &- abw^a\tilde{f}\end{aligned}\tag{61}$$

Grouping like terms and dividing by w^a we see

$$w^2(w-1)\frac{d^2\tilde{f}}{dw^2} + w[w(2a+2-c) + (b-a-1)]\frac{d\tilde{f}}{dw} + a(a-c+1)w\tilde{f} = 0 \quad 62.$$

Dividing by $-w$ to reduce order and get it into correct form,

$$w(1-w)\frac{d^2\tilde{f}}{dw^2} + [-w(2a+2-c) + (1+a-b)]\frac{d\tilde{f}}{dw} + a(a-c+1)w\tilde{f} = 0 \quad 63.$$

This is $\text{HG}_{a, a-c+1, 1+a-b}(\tilde{f}) = 0$. Thus

$$\tilde{f}(w) = F(a, a-c+1, a-b+1; w) \quad 64.$$

. Thus

$$F_\infty(z) = z^{-a}F\left(a, a-c+1, a-b+1; \frac{1}{z}\right) \quad 65.$$

and by symmetry,

$$G_\infty(z) = z^{-b}F\left(b, b-c+1, b-a+1; \frac{1}{z}\right) \quad 66.$$

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